Dang Phuong Nam

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Education

FPT University — Hanoi, Vietnam
Bachelor of Artificial Intelligence — *Oct 2023 – Dec 2027 (Expected)*

Publications

Conflict-Aware RAG Routing: Balancing Cost and Accuracy via Context Contradiction 2026

- **Dang Phuong Nam**, Tran Huy Tuan, Doan Duy Long, Cao Van Mai. *13th IEEE International Conference on Intelligent Systems (IEEE IS'26).* **First author**; accepted & presented.

Probabilistic Modeling of Stock Breakout Success during Consolidation Phases in the Vietnamese Equity Market 2026

- Doan Duy Long, Tran Huy Tuan, **Dang Phuong Nam**, Cao Van Mai. *11th International Conference on Information Technology and Digital Applications (ICITDA 2026).* Accepted.

Experience

Full-stack Developer Intern — Aug 2025 – Dec 2025
XTEK — *Hanoi, Vietnam*

- Learned full-stack web development through mentored internship projects using React.js and Spring Boot
- Built the frontend of a shoe e-commerce website with React.js
- Developed the backend of a student management system with Spring Boot

Projects

Conflict-Aware RAG Routing | *First Author* | *Python, PyTorch, Hugging Face, scikit-learn* 2026

- Designed a bidirectional NLI context-conflict score (DeBERTa-v3 cross-encoder) and a 5-feature Random Forest router deciding between a local Qwen2.5-1.5B and the DeepSeek-V4-Flash API
- Built leakage-resistant nested 5-fold budget calibration; evaluated on 3,000 multi-hop questions
- Saved **∼35% of API calls** while retaining **83.2–92.2% of Always-LLM F1**; **+14.35 F1** over RRF-only routing in the high-conflict HotpotQA subgroup ($p = 0.006$)

Multi-Task Stock Transformer | *Capstone, Team Leader* | *PyTorch, Qwen2.5, PatchTST, QLoRA* 2026 – Present

- Leading the design of a shared decoder-only SLM (Qwen2.5-0.5B) that fuses PatchTST price patches, financial text and ESG features for trend classification and generated rationale
- Proposed an NLI-based consistency loss that penalizes rationales contradicting the predicted trend

VietSign Edge AI | *YOLOv8n, MobileNetV3, ByteTrack, TFLite, Android CameraX* 2025

- Built a two-stage on-device pipeline: YOLOv8n sign detection followed by MobileNetV3-Small classification of 48 Vietnamese traffic sign classes (QCVN 41:2019)
- Added ByteTrack tracking and INT8 TFLite quantization for offline real-time inference on Android

Verba | *Full-Stack Engineer* | *FastAPI, PostgreSQL, PhoWhisper, Gemini* Jan 2026 – Apr 2026

- Users upload meeting audio and get transcripts summarized into key points, decisions, tasks and deadlines
- Integrated PhoWhisper/faster-whisper STT, Gemini summarization, synced audio player and .docx export

Technical Skills & Certifications

Languages: Python, Java, TypeScript / JavaScript, SQL
ML & Deep Learning: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, QLoRA / PEFT
LLM & RAG: Hybrid retrieval (BM25 + dense, RRF), NLI cross-encoders, LLM APIs (DeepSeek, Gemini), Whisper
Computer Vision & Edge: YOLOv8, MobileNetV3, OpenCV, TFLite (INT8)
Backend & Tools: FastAPI, Spring Boot, React.js, PostgreSQL / pgvector, Docker, Git, Linux
Certifications: NLP Spec. (DeepLearning.AI), Reinforcement Learning Spec. (U. of Alberta), Fundamentals of Deep Learning (NVIDIA), AI Enterprise Workflow (IBM)